

# Chess Game Analyzer, an introduction

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# Contents

|          |                                                              |           |
|----------|--------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                          | <b>7</b>  |
| 1.1      | Computer chess for humans . . . . .                          | 7         |
| 1.2      | The five core metrics . . . . .                              | 9         |
| 1.2.1    | Data model and metric plots . . . . .                        | 9         |
| 1.2.2    | Space . . . . .                                              | 10        |
| 1.2.3    | Mobility . . . . .                                           | 10        |
| 1.2.4    | King Safety . . . . .                                        | 10        |
| 1.2.5    | Threats . . . . .                                            | 11        |
| 1.2.6    | The Fireteam Index signature . . . . .                       | 11        |
| 1.2.7    | Balance . . . . .                                            | 11        |
| <b>2</b> | <b>Examples</b>                                              | <b>13</b> |
| 2.1      | Estrin-Berliner, 1965 . . . . .                              | 13        |
| 2.1.1    | Opening phase (Moves 1–5): The Two Knights Defense . . . . . | 14        |
| 2.1.2    | The critical opening battle (Moves 6–15) . . . . .           | 14        |
| 2.1.3    | The blunder . . . . .                                        | 14        |
| 2.1.4    | The tactical middlegame (Moves 15–28) . . . . .              | 15        |
| 2.1.5    | The endgame, initial phase (Moves 28–34) . . . . .           | 16        |
| 2.1.6    | The final phase (Moves 35–42) . . . . .                      | 17        |
| 2.1.7    | FTI summary . . . . .                                        | 17        |
| 2.2      | Carlsen-Caruana, Game 6, 2018 . . . . .                      | 21        |
| 2.2.1    | The early opening moves (1-4) . . . . .                      | 21        |
| 2.2.2    | The opening phase (5-15) . . . . .                           | 22        |
| 2.2.3    | The early middlegame (16-25) . . . . .                       | 23        |
| 2.2.4    | The later middlegame (26-33) . . . . .                       | 24        |
| 2.2.5    | The early endgame (34-67) . . . . .                          | 25        |
| 2.2.6    | The later endgame (67-80) . . . . .                          | 27        |
| 2.2.7    | FTI summary . . . . .                                        | 29        |
| <b>3</b> | <b>Appendix: Chess Game Analyzer module reference</b>        | <b>33</b> |
| 3.1      | Overview . . . . .                                           | 33        |
| 3.2      | Background for Newcomers . . . . .                           | 33        |
| 3.2.1    | What is Stockfish? . . . . .                                 | 33        |
| 3.2.2    | What is a Centipawn? . . . . .                               | 34        |
| 3.2.3    | Python Requirements . . . . .                                | 34        |

|        |                                                        |    |
|--------|--------------------------------------------------------|----|
| 3.3    | Installation . . . . .                                 | 34 |
| 3.4    | Quick Start . . . . .                                  | 35 |
| 3.4.1  | Command Line . . . . .                                 | 35 |
| 3.4.2  | Python API . . . . .                                   | 35 |
| 3.5    | Positional Metrics Explained . . . . .                 | 36 |
| 3.5.1  | Space . . . . .                                        | 36 |
| 3.5.2  | Mobility . . . . .                                     | 36 |
| 3.5.3  | King Safety . . . . .                                  | 37 |
| 3.5.4  | Threats . . . . .                                      | 37 |
| 3.6    | The Fireteam Index (FTI) . . . . .                     | 38 |
| 3.6.1  | Background: Chess Metrics and Win Prediction . . . . . | 38 |
| 3.6.2  | FTI Signatures . . . . .                               | 38 |
| 3.6.3  | Prediction Algorithms . . . . .                        | 38 |
| 3.7    | Concrete Example: Estrin vs Berliner (1965) . . . . .  | 39 |
| 3.7.1  | The Game (C57 – Two Knights Defense) . . . . .         | 39 |
| 3.7.2  | Analysis Output . . . . .                              | 39 |
| 3.7.3  | Positional Metrics Summary . . . . .                   | 40 |
| 3.7.4  | Move-by-Move Data Extract . . . . .                    | 40 |
| 3.8    | Command Line Interface . . . . .                       | 40 |
| 3.8.1  | Core Options . . . . .                                 | 41 |
| 3.8.2  | Report Options . . . . .                               | 41 |
| 3.8.3  | Fireteam Index Prediction (Single Game) . . . . .      | 41 |
| 3.8.4  | Book Mode (Multiple Games) . . . . .                   | 41 |
| 3.8.5  | Examples . . . . .                                     | 41 |
| 3.9    | Python API . . . . .                                   | 42 |
| 3.9.1  | Main Classes . . . . .                                 | 42 |
| 3.9.2  | Utility Functions . . . . .                            | 43 |
| 3.10   | Parameter Reference . . . . .                          | 44 |
| 3.10.1 | Analysis Parameters . . . . .                          | 44 |
| 3.10.2 | FTI Prediction Parameters . . . . .                    | 44 |
| 3.10.3 | Available Weight Presets . . . . .                     | 44 |
| 3.10.4 | Move Classification Thresholds . . . . .               | 44 |
| 3.11   | Output Formats . . . . .                               | 45 |
| 3.11.1 | L <sup>A</sup> T <sub>E</sub> X Report . . . . .       | 45 |
| 3.11.2 | JSON Output . . . . .                                  | 45 |
| 3.11.3 | Raw Positional Data . . . . .                          | 46 |
| 3.12   | License . . . . .                                      | 46 |
| 3.13   | Acknowledgments . . . . .                              | 46 |
| 4      | References . . . . .                                   | 47 |

# Preface

This is a guide of how to use a program I wrote to analyze chess games you input (as a pgn file on your computer) called the *chess game analyzer*, or *CGA* [CGA26]. The program uses other chess programs to output a math-dense LaTeX file, called a *Quantitative-Analytic Report on Chess* (or QARC), where the results from those other programs are collected in a human-readable format. Full disclosure: For creating CGA, I did get a lot of help from AI: **gemini** (Google) for chess-theoretic questions, **chatGPT** (openAI) for outline ideas, and **claude** (Anthropic) for Python questions.

The CGA is a Python module providing an advanced chess analysis pipeline that goes beyond piece evaluations. It combines **Stockfish** engine analysis with “home-made” positional metrics to generate quantitative LaTeX reports on chess games, machine-readable data, and win-prediction metrics. In addition to generating educational content, it collects useful game data into tables, and computes “Fireteam Index” (or FTI, defined below) signatures. As we will see below, the FTI is a family of metrics inspired by the medieval board game rithmomachia [Ri25] to measure sustained positional dominance.

In fact, all this, the CGA module and related programs<sup>1</sup>, was inspired by the “territorial” approach that proper victories take in rithmomachia. Can such an approach be helpful in chess analysis? Maybe more refinements to CGA are needed before I can answer that, but that question is the main motivation for this work.

Another goal was to create “Prediction Algorithms” for chess games in the hope of predicting a win during live play. Of course, that is impossible in general, but the question remains: what can be done, provided we allow some (not too many) inaccuracies? To address this, CGA implements three algorithms that use the FTI to predict game outcomes:

- **Per-Ply.**
  - Examines FTI at each ply (=half-move) after ply 16.
  - Predicts *win* if  $\text{FTI} > \epsilon$  (default  $\epsilon = 0.2$ ) for 10 consecutive plies.
  - Added plus: Captures sudden, decisive advantages.
- **Windowed**
  - This is a “smoothed” version of Ply.
  - Computes a 10-ply rolling average of FTI,  $\text{FTI}_{10}$  say.

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<sup>1</sup>For instance, CGA has an add-on that creates animated gifs of games, but such add-ons will be discussed separately.

- Applies the same streak detection to smoothed values: Predicts *win* if  $\text{FTI}_{10} > \epsilon$  for 10 consecutive plies.
- Filters out tactical noise; captures sustained trends.
- **Balance**
  - Counts plies where  $\text{FTI} > \epsilon$  versus  $\text{FTI} < -\epsilon$
  - Measures volatility (sign changes in the difference between these counts)
  - Computes confidence =  $|B_n| \cdot (1 - V/V_{\max})$
  - Predicts *draw* if confidence  $< 0.15$ ; otherwise *win/loss* (that is, not a draw).
  - Best at detecting drawn games and balanced struggles

The example games are well-known and well-analyzed, so you can compare CGA analysis here with the GM analysis published already in books, or blogs or on youtube. The first one is a World Correspondence Chess game played in 1965, Estrin-Berliner, §2.1. The second is Game 6 of the World Chess Championship match Carlsen-Caruana in London, Nov. 2018, §2.2.

You don't need to know how to use **Stockfish** or **Python** in order to run the CGA programs. However, you do need to know how to install them on your computer if they aren't already. Installation help and the syntax for using CGA is presented in detail in the appendix, ch. 3. Don't worry about the syntax or installing software yet. First, let's learn a little more about the CGA module.

# Chapter 1

## Introduction

As mentioned, detailed reference for the CGA module is given in an appendix, ch. 3. However, the basic idea is that the CGA module takes a pgn file as input. The output consists of a latex report which contains the following information about the game (or games) in the pgn file.

- Move-by-move positional analysis with Space, Mobility, King Safety, and Threats metrics
- **Stockfish** suggestions of playable alternative moves within 50 centipawns of the best
- Game character classification (balanced/tense/tactical/chaotic, one-sided/seesaw)
- Brilliant sacrifice detection with sound/speculative classification
- Five different Fireteam Index metrics capturing different playing styles
- Other metrics, such as Balance and Volatility, measure likelihood of a draw.
- Three prediction algorithms (Per-Ply, Windowed, Balance) for win or draw predictions.
- Publication-quality LaTeX output with chess diagrams, plots, and methodology
- Book mode for analyzing multi-game PGN files with aggregate statistics
- Raw data export for downstream computational analysis

These will be explained, with examples, below.

### 1.1 Computer chess for humans

When a human plays chess in a given position, they use both logic (to trace through potential moves and the ramifications of their opponent's responses) and experience (to restrict their analysis to a small number of moves) in their effort to win. The human player makes a move based on rigorous reasoning and their previous experience with games similar to the one they are currently playing. They pick one that, they feel, gives them the best position

no matter what their opponent does in response. Of course, a player will not want to lose any advantage they feel they have in the position, whether material, or space, or mobility, or king safety, or potential threats they can mount. At the same time, they want their move to also restrict their opponent’s advantages in the position.

A computer plays chess using a different process. The computer player also uses logic to sift through legal moves for the best one. The difference is that the computer can, with the help of a chess engine like **Stockfish**, evaluate positions very rapidly and they pick the move that both maximizes their own position evaluation and minimizes the evaluations of each response by their opponent. In other words, to a computer, playing chess is analogous to solving a math problem. Can humans learn a lesson or two from how the computer evaluates positions? Let’s look more closely at this algorithmic perspective the computer player takes.

A real-valued function  $m : \text{FEN} \rightarrow \mathbb{R}$  that takes in a position of a chess game and returns a positive number is called a *chess metric*. Here, **FEN** simply denotes the collection of all possible FEN positions<sup>1</sup> and  $\mathbb{R}$  is the standard notation for the set of real numbers.

One easy metric all players use is the *piece value metric* for White and for Black:  $PV_W(\text{fen})$  denotes the total value of the white pieces in the position *fen*, and  $PV_B(\text{fen})$  using the standard values for the Black pieces in the give position. Most players use the usual values<sup>2</sup> for the pieces:

| piece  | value    |
|--------|----------|
| king   | $\infty$ |
| queen  | 9        |
| rook   | 5        |
| bishop | 3        |
| knight | 3        |
| pawn   | 1        |

The reader can check that, in the game’s starting position, we have

$$P_W(\text{start}) = P_B(\text{start}) = 39.$$

Another example is a chess metric we’ll denote by **eval**, the evaluation metric from the chess engine **Stockfish**. This positional metric measures advantages (including piece values) in *centipawns* (cp) — hundredths of a pawn. An evaluation of +100 (that is,  $\text{eval}(\text{fen}) > 100$ ) means White is ahead in the given **FEN** position by the positional equivalent of roughly one pawn. An evaluation of −250 means Black has about a 2.5 pawn advantage.

---

<sup>1</sup>FEN stands for Forsyth–Edwards Notation, a standard notation for describing a board position, typically represented as an alphanumeric list or string, see [FEN]. Forsyth was a Scottish journalist from the 1800s while Edwards was an American computer scientist active in computer chess from the late 1900s to early 2000s.

<sup>2</sup>For more background on the different ways to evaluate chess pieces, and there are quite a lot of them, see the wikipedia article [PV] and the references there (including references to Berliner’s book [Be99]).



| Evaluation             | Meaning             |
|------------------------|---------------------|
| 0 to $\pm 50$          | Roughly equal       |
| $\pm 50$ to $\pm 150$  | Slight advantage    |
| $\pm 150$ to $\pm 300$ | Clear advantage     |
| $\pm 300$ to $\pm 500$ | Winning advantage   |
| $\pm 500+$             | Decisive advantage  |
| $\pm 10000$            | Checkmate is forced |

The modern **Stockfish** versions primarily uses NNUE (neural network) evaluation to compute **eval**. Unfortunately, that metric is a bit of a black box. It appears to be impossible, or at least very impractical, to directly extract from the modern **Stockfish** other, more “territorial” metrics such as Space.

Let’s dive into that topic next.

## 1.2 The five core metrics

The **Stockfish** evaluation **eval** tells the story, metrics explain why.

As mentioned, the modern **Stockfish** **eval** is a bit of a black box.

Given a chess position, who do we think of it from a territorial perspective? These natural questions arise:

- How to we compute the “space” controlled by a given side?
- How to we compute the “mobility” of pieces in the position?
- How do we compute the safety of the king? This is, once it’s defined, a defensive metric.
- How do we compute the “threats” one side has against their opponent? Conversely, once it’s defined, this is an offensive metric.

The four metrics Space/Mobility/King Safety/Threats use the Python module **python-chess** to parse the position and then Python to directly compute the metrics based on reasonable formulas<sup>3</sup>. These formulas are designed to address these questions.

### 1.2.1 Data model and metric plots

Raw positional data fields are printied to the bottom of the latex QARC file (but does not appear in the compiled pdf). CGA uses this data to plot various chess metrics.

The plots of the metrics are either Per-ply or Windowed.

- Per-ply: Looks for 10 consecutive positions where one side dominates (FTI > 0.20).
- Windowed/Smoothed: Uses a 10-ply rolling average to filter out tactical noise.

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<sup>3</sup>For details on **python-chess** and the formulas for the metrics (Space, Mobility, and so on), please see the following subsections or the appendix, chapter 3.

### 1.2.2 Space

“How much ‘room’ do your pawns give your pieces to move in the center?”

Space evaluation counts squares in the center (files c-f, ranks 2-4 for White, 5-7 for Black) that are:

1. Attacked by at least one friendly pawn, *and*
2. Not occupied by any enemy piece

The computation is weighted by the number of pieces behind the pawn chain — this rewards positions where you control territory *and* have pieces active in that space. The Space term is scaled by total piece count; it matters more in closed positions with many pieces.

Space is a middlegame metric — it becomes near-meaningless in endgames when most pieces are traded.

### 1.2.3 Mobility

“How many useful squares can your pieces actually land on? (More squares = more power)”

Mobility counts legal moves for each piece type, with different weights:

- **Knights:** Safe squares attacked (excluding squares defended by enemy pawns)
- **Bishops:** Safe squares on diagonals (bonus for long, unobstructed diagonals)
- **Rooks:** Squares on files/ranks (bonus for open files, 7th rank penetration)
- **Queens:** Combination of bishop and rook mobility patterns

“Safe squares” exclude those defended by enemy pawns or where exchanges would lose material. Mobility in the opponent’s territory is valued higher than mobility in your own.

Mobility tracks piece activity. Watch it climb during development and converge as long-range pieces (bishops, rooks, queens) are trade off.

### 1.2.4 King Safety

“How well is your King shielded by pawns, and how many enemy ‘attack units’ are nearby?”

King safety combines several factors:

- **Pawn shield:** Pawns on 2nd/3rd rank directly in front of the castled king
- **King geometry:** Distance of enemy pieces to your king.
- **Attack units:** Count of enemy pieces attacking squares adjacent to your king
- **Safe checks:** Number of safe checking squares available to the opponent

King Safety can be positive and negative. Negative values in the middlegame mean danger. Negative values in the endgame can mean active, fighting kings.

### 1.2.5 Threats

“Are you hanging pieces, or are your small pieces attacking the opponent’s big pieces?”

Threat evaluation considers:

- Hanging pieces (attacked but not adequately defended)
- Safe pawn advances that create threats
- Minor pieces attacking major pieces
- Weak squares in the pawn structure

Threats measures potential piece captures. High threat values show tension; when they drop sharply after exchanges, material or positional concessions have been made.

**Example 1.2.1.** *For example, the  $-319cp$  blunder at move 15 in §2.1.2 below shows what happened but the Mobility and Threats changes in moves 8-14 show why the position was ripe for sharp play from both sides.*

### 1.2.6 The Fireteam Index signature

This is a linear combination of those four metrics above.

The Fireteam Index is a composite metric inspired by the ancient mathematical board game rithmomachia, where a “progression victory” is achieved when a coordinated group of pieces (a “fireteam”) establishes control deep in enemy territory [Ri25]. Similarly, in chess, sustained positional dominance across multiple factors often precedes victory.

The FTI weights can be thought of as *signatures* that capture different playing styles. By adjusting which positional factors receive emphasis, the FTI can be tuned to better predict outcomes for players with different strategic preferences.

### 1.2.7 Balance

Balance is the key draw-detection metric. It combines:

- Normalized Balance ( $B_n$ ): What fraction of positions favored each side
- Volatility ( $V$ ): How often the advantage changed hands
- Confidence ( $C$ ):  $C = B_n \cdot (1 - V/V_{\max})$ , where  $V_{\max}$  is the maximum volatility for that game.

The idea is that if the positional metrics are roughly equal for most of the game then neither side dominated the Space/Mobility/Threats picture consistently. Similarly, high volatility means the advantage kept changing hands during the course of the game. The *Balance algorithm* will predict the game ends in a draw if  $|B_n| < 0.2$  or  $V \geq 10$ . This data is part of the QARC output.



# Chapter 2

## Examples

The first example game, ironically, was won by Hans Berliner, a highly respected computer pioneer and computer science researcher.

### 2.1 Estrin-Berliner, 1965

This is a popular game for commentary and analysis. The point here is not to try to provide a summary of the analysis of others but to report on what CGA says about these games.

The QARC file on the game Estrin vs. Berliner (1965) describes it as a “chaotic seesaw battle,” where the advantage changed hands multiple times [Be65]. We’ll walk you through this game using the provided metrics, you can use them to explain the dynamics of the positional components (Space, Mobility, King Safety, and Threats).

White, played by the Russian chess player Yakov Estrin, took joint first place in the USSR Correspondence Championship a few years earlier, in 1962. Estrin is best known for being the seventh ICCF World Champion a few years later, in 1972–1976.

Black, played by American Hans Berliner, was also a well-known chess player. Here is a quotation from Dunne’s remembrance article for Has Berliner written for **Chess Life** [Du17]:

The finals of the Fifth World Correspondence Chess Championship produced a score of 14 – 2, 12 wins and four draws, a three point margin over the second place finisher, a spread unmatched in World Championship play. In that final, Hans also played what is widely regarded as the greatest correspondence game ever played, and listed in Andy Soltis’ book **The 100 Best Chess Games of the 20th Century** as the top game of that tumultuous century. It is a game that has resisted major improvement on Black’s part for over half a century.

In fact, that last statement of Dunne’s is, thanks to progress in chess engines such as **Stockfish**, no longer completely true. However, we aren’t going down that rabbit hole in this guide.

### 2.1.1 Opening phase (Moves 1–5): The Two Knights Defense

With that background out of the way, let's see what the CGA had to say about this game.

**1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6 4.Ng5 d5 5.exd5**

This is the famous Two Knights Defense with the Fried Liver Attack. Estrin plays 4.Ng5, immediately creating tension. Let's see how the metrics capture this.

Notice how Space remains roughly equal at 0.40 for both sides through move 9. Space measures control of central squares backed by your pawns — essentially, how much room your pieces have to maneuver. After

**5.exd5**

White captures a pawn but Black will strike back. Watch what happens after 5...b5 — the Ulvestad Variation. The usual move in this position is 5. ... Na5.

Data at ply 10/move 5: Eval: +0.45

Look at the Threats metric — White's threats spike to 10.80 because the b5-pawn is hanging. But if you compare Black's Mobility to White's, you see Berliner is accepting material deficit for piece activity.

### 2.1.2 The critical opening battle (Moves 6–15)

**6. Bf1 Nd4 7. c3! Nxd5! 8. Ne4?! Qh4?**

From the game's annotation by *Stockfish*, both 8 ... Nc6 and 8 ... Ne6 are preferred.

**9. Ng3! Bg4?**

Again, *Stockfish* recommends 9 ... Nc6.

Data at ply 18/move 9:

Eval: +1.55

The game goes on (see [Be65]):

**10. f3! e4?**

Black loses 67cp. Consider instead: Bd7, Be6, Bd6.

**11. cxd4 Bd6! 12. Bxb5+ Kd8 13. O-O exf3 14. Rxf3! Rb8**

This sets up a crucial position.

### 2.1.3 The blunder

With threats from both sides, the next move is important.

**15. Be2??**

*Looks* good but, in fact, some commentators say this natural-looking move is the move that lost the game. *Stockfish* recommends 15 Bf1 or else 15 a4.

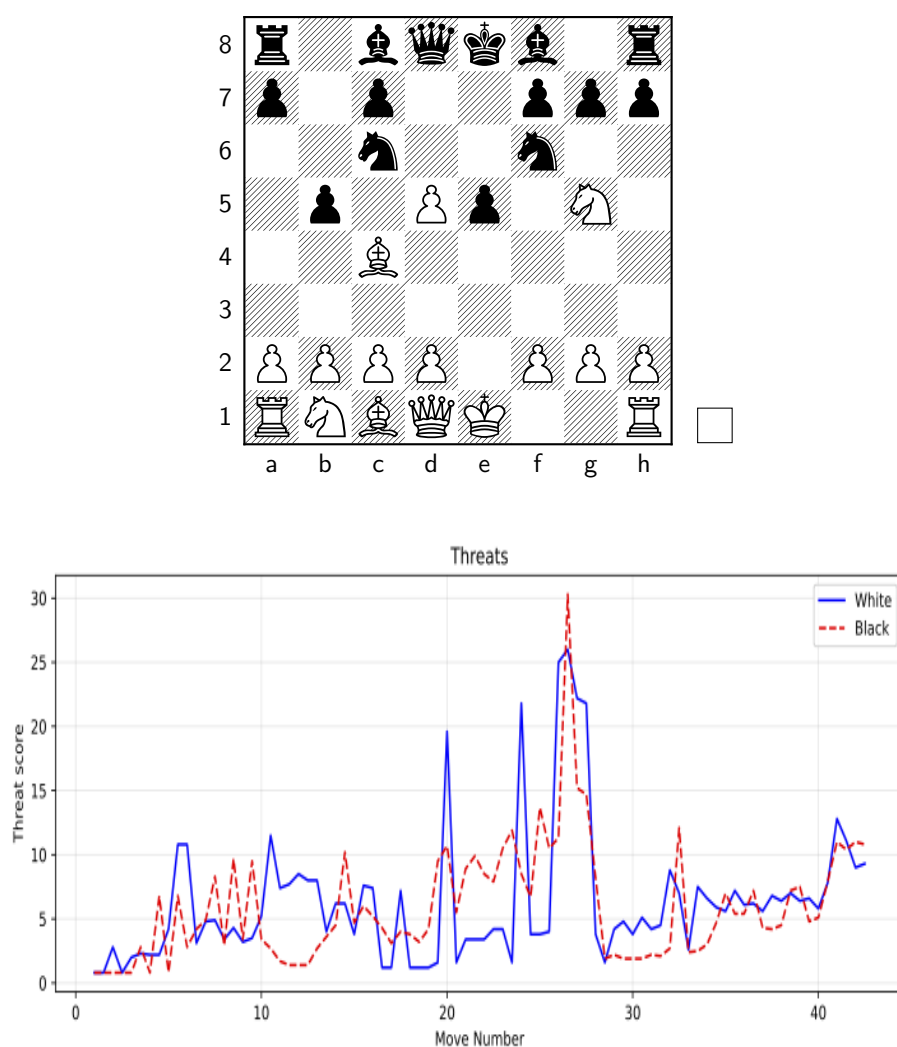


Figure 2.1: (top) Position after 5 ... b5 in Estrin-Berliner (1965). (bottom) The Threats metric for that game.

Data at ply 31:

Eval: -1.26

### 2.1.4 The tactical middlegame (Moves 15–28)

The game continues:

16. Bxf3! Qxd4+?

Stockfish preferred instead 16... Re8.

17. Kh1! Bxg3! 18. hxg3! Rb6 19. d3! Ne3! 20. Bxe3! Qxe3! 21. Bg4 h5 22. Bh3! g5 23. Nd2! g4! 24. Nc4! Qxg3! 25. Nxb6! gxh3! 26. Qf3 hxg2+ 27.



Figure 2.2: (top) Position after 9 ... Bg4 in Estrin-Berliner (1965). (bottom) The Mobility metric for that game.

**Qxg2! Qxg2+**

The queens are trading off now.

### 2.1.5 The endgame, initial phase (Moves 28–34)

Without long-range pieces, the Mobility metric drops. We enter a new phase of the endgame.

**28. Kxg2 cxb6! 29. Rf1 Ke7! 30. Re1+! Kd6 31. Rf1! Rc8?**

Stockfish recommends Ke6, Ke7, or Rh7 instead.

Data at ply 62/move 31:



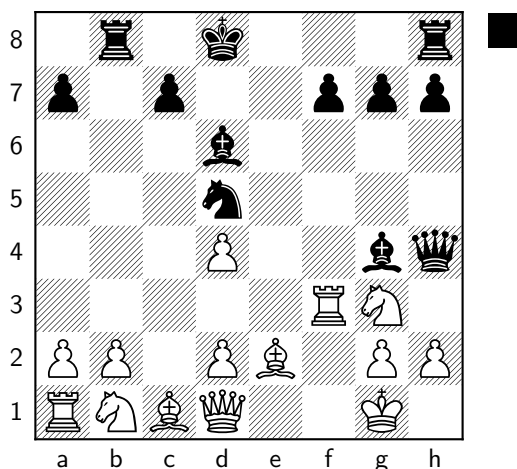


Figure 2.3: Position after 15 Be2 in Estrin-Berliner (1965).

Eval: -1.21

**32. Rxf7?! Rc7! 33. Rf2?**

Stockfish suggests to consider instead the move Rf6+.

**33...Ke5 34. a4??**

This is a more serious error, but not of the same magnitude as 15 Be2??. With 34. a4??, eval drops from -2.24 to -4.01. Here, *Stockfish* recommends instead Rf8.

**34...Kd4! 35. a5?! Kxd3**

### 2.1.6 The final phase (Moves 35–42)

The final moves of the game.

**36. Rf3+ Kc2?! 37. b4 b5! 38. a6?! Rc4?! 39. Rf7 Rxb4 40. Rb7?! Rg4+! 41. Kf3?**

Stockfish suggests instead Kh3 or Kf3.

**41...b4?! 42. Rxa7! b3! 0-1**

### 2.1.7 FTI summary

The various types of FTIs are designed to characterize the type of play. Recall that the FTI metric is a linear combination of the four positional metrics, Space (S), Mobility (M), King Safety (KS), and Threats (T):

$$FTI = w_S S + w_M M + w_{KS} KS + w_T T.$$

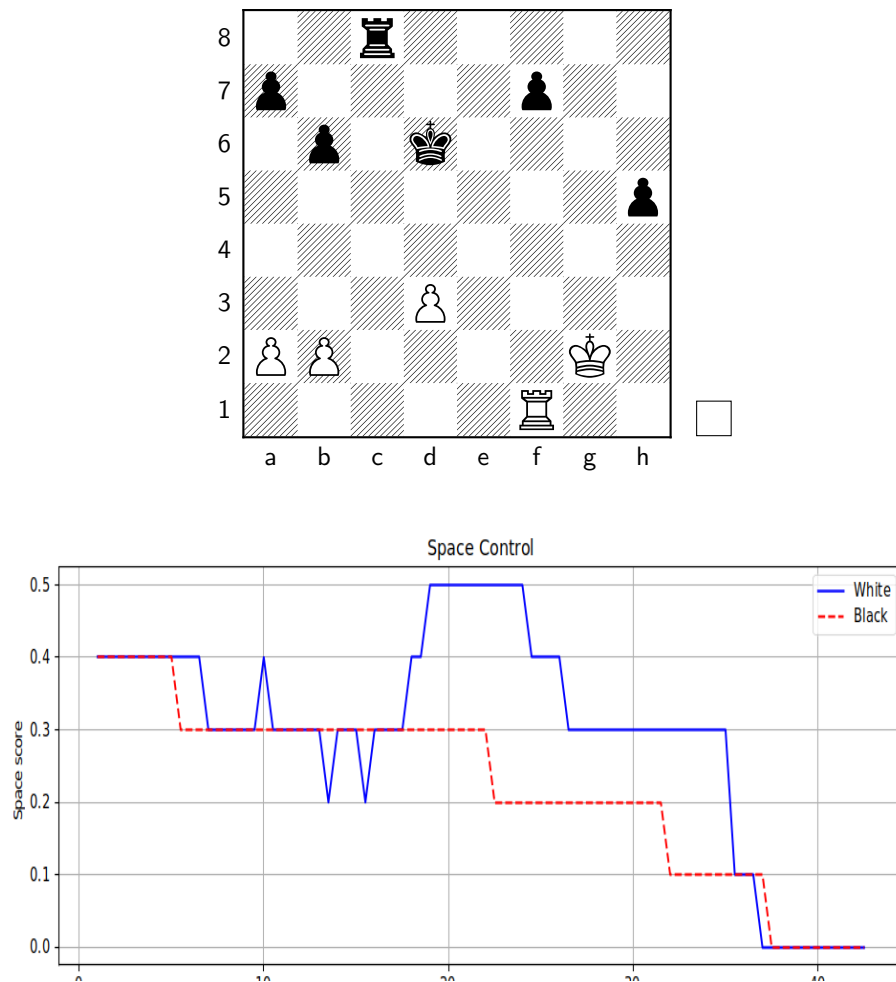


Figure 2.4: (top) Position after 31 ... Rc8? in Estrin-Berliner (1965). (bottom) The Space metric for that game.

Think of the FTI as a coordinated pressure gauge. Using it to predict a game's outcome requires a player to hold positional dominance for 10 consecutive 'plies' (half-moves) to predict a win. For the Estrin-Berliner game, predicting the outcome (win/loss/draw) using Fireteam Indices is a struggle because the game was so volatile.

In this case, CGA was run over all the games Berliner played for the 1965 World Correspondence Chess championship. The CGA program computes the weights ( $w_S, w_M, w_{KS}, w_T$ ) such that the FTI with those "signature" weights, the plot predicts the outcomes more successfully than any other choices. The outcome was

$$(w_S, w_M, w_{KS}, w_T) = (0.319, 0.675, 0.003, 0.003),$$

with  $R^2 = -12.525$  (indicating a poor fit).

Either by looking at the plots for FTI3 and FTI5 or that signature FTI plot, we see CGA suggests Berliner has a solid, strategic style.

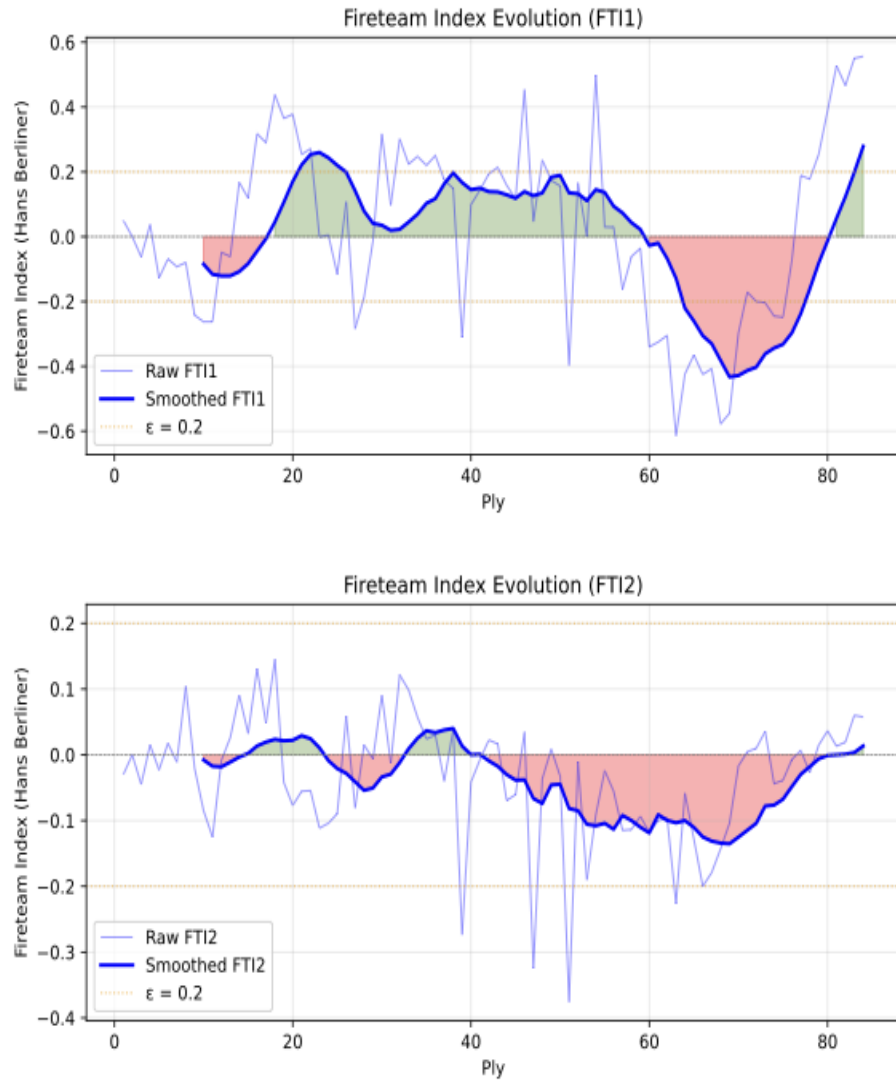


Figure 2.5: Both plots are for Estrin-Berliner, 1965. (top) Fireteam index, FTI1 (harmonious). (bottom) Fireteam index, FTI2 (tactical). The second plot never exceeds the confidence threshold of  $\pm 0.2$ .

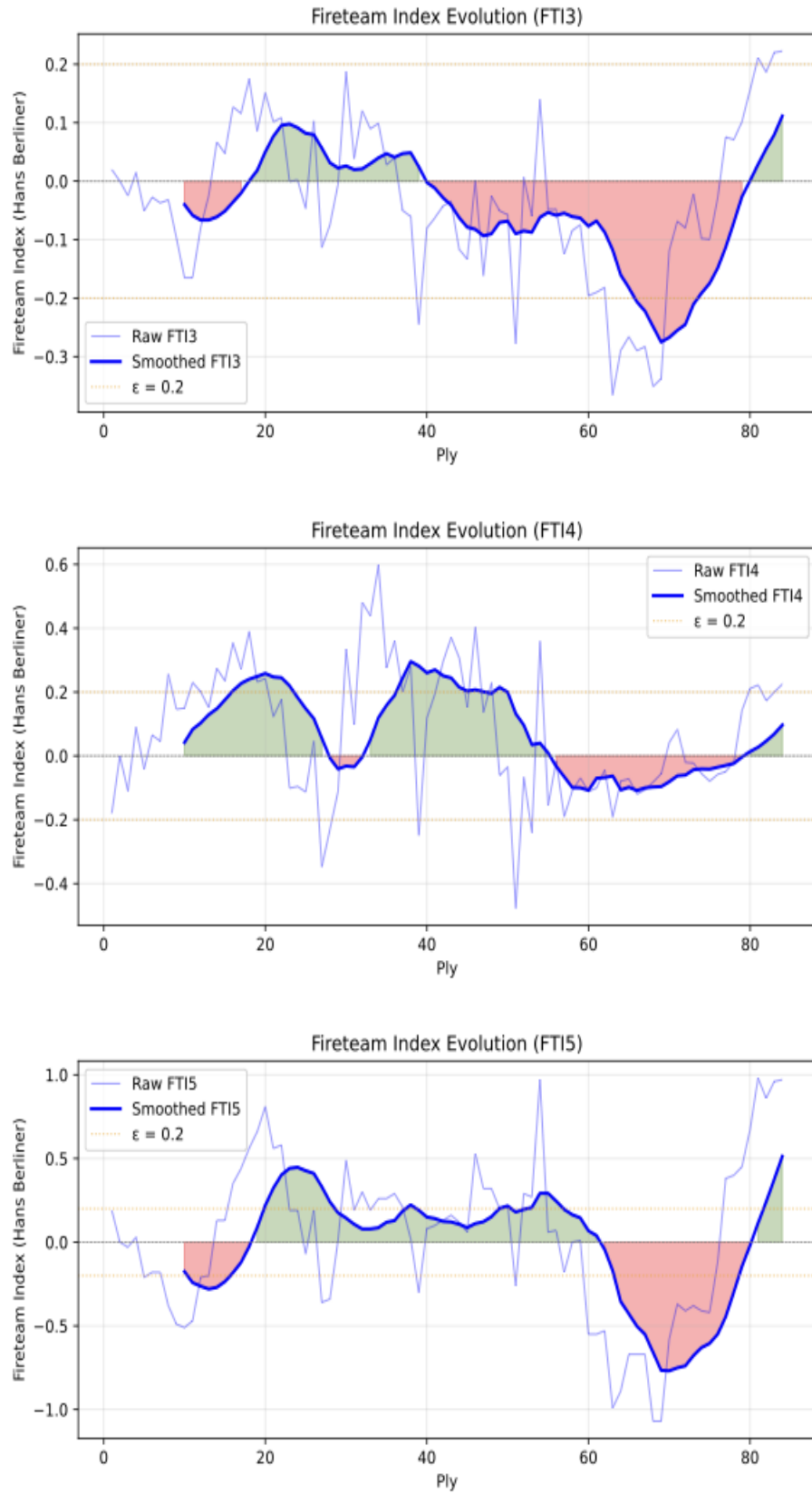


Figure 2.6: These plots are also for Estrin-Berliner, 1965. (top) Fireteam index, FTI3 (strategic). (middle) Fireteam index, FTI4 (dynamic). (bottom) Fireteam index, FTI5 (solid).

Of all the FTI plots in Figures 2.5 and 2.6, the last one is the one that most strongly suggests a win for Black. FTI1 and FTI3 also do suggest a Black win, but with less confidence.

## 2.2 Carlsen-Caruana, Game 6, 2018

At this point in the match, the players were tied 2.5-2.5, with 5 draws in a row. This game was another draw but exciting none-the-less. In fact, on move 67 it appears Caruana missed a win.

The QARC file on the 12 games in the Carlsen-Caruana match, London (November 2018) is available online as a pdf [CC18]. We'll walk you through this game using the provided metrics, you can use them to explain the dynamics of the positional components (Space, Mobility, King Safety, and Threats).

Of course, both players are well-known. In 2018, the 27-year-old Norwegian was the undisputed king of chess, having held the World Championship title since 2013 and the world No. 1 ranking since 2011. The 26-year-old American-Italian grandmaster, in 2018, became the first American to challenge for the undisputed world chess title since Bobby Fischer in 1972. At the time of their match, their ELO ratings were nearly the same.

### 2.2.1 The early opening moves (1-4)

Look at the plot for **Stockfish**'s eval function (see Figure 2.7).

Notice how the line hovers around zero for the first 10-15 moves, then drifts into negative territory, favoring Caruana. The game reaches its most critical point around moves 67-68 where we see a dramatic spike to about  $-2.5$  pawns, a huge advantage for Black. Unfortunately for Caruana, that was followed by an equally dramatic swing back toward zero.

**1. e4 e5 2. Nf3 Nf6 3. Nxe5 d6 4. Nd3?!**

While this is part of opening theory<sup>1</sup>, **Stockfish** registered Carlsen's move as an inaccuracy, preferring instead Nf3 or Nc4.

**Position 1: After 4. Nd3**

Biggest evaluation swing(s) (39cp)

Evaluation: +0.00

---

<sup>1</sup>ECO C42: Petrov's Defence, Karklins-Martinovsky Variation, to be precise.

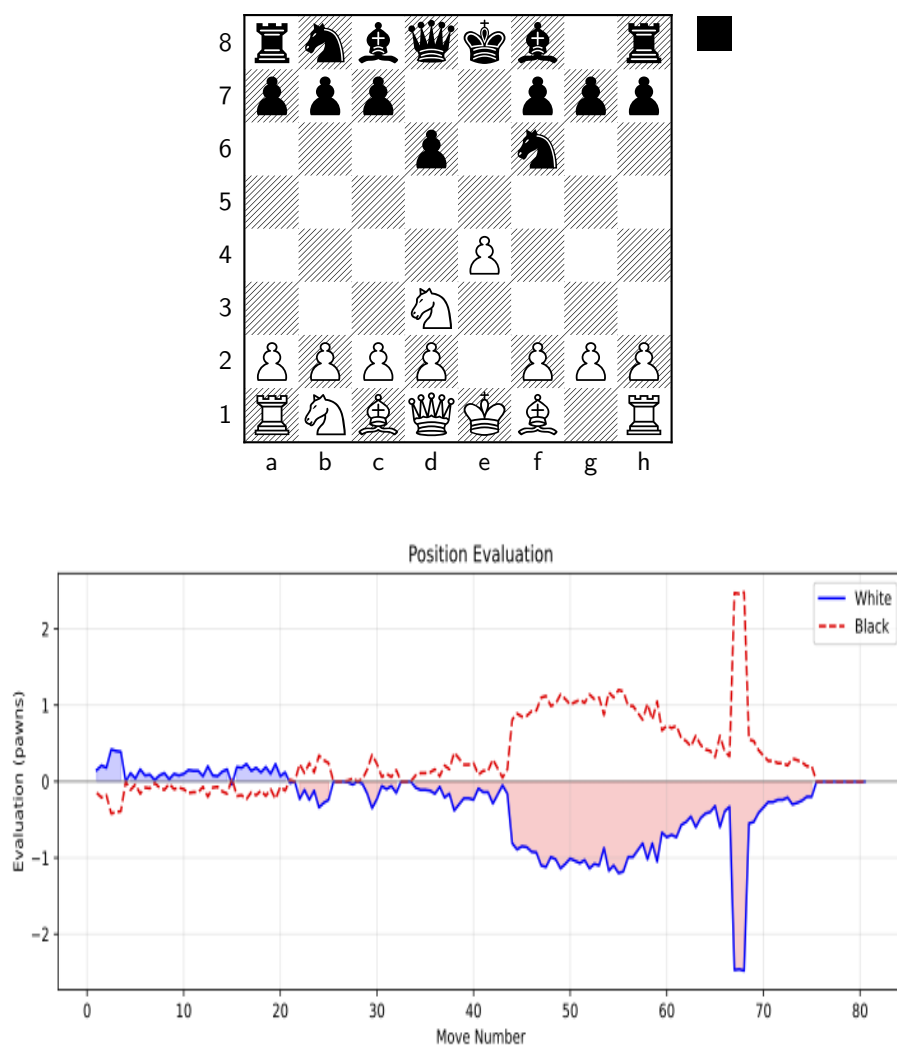


Figure 2.7: (top) Position after 4 Nd3?! in Carlsen-Caruana, Game 6 (2018). (bottom) The eval metric for that game.

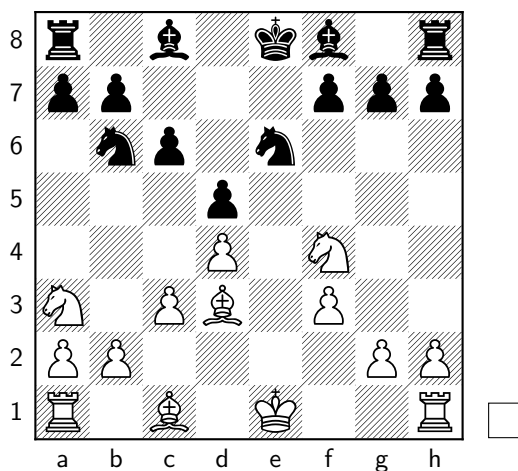
### 2.2.2 The opening phase (5-15)

4 ... Nxe4 5. Qe2 Qe7 6. Nf4 Nc6 7. Nd5 Nd4! 8. Nxe7! Nxe2! 9. Nd5! Nd4!  
10. Na3 Ne6 11. f3! N4c5! 12. d4 Nd7 13. c3 c6! 14. Nf4! Nb6! 15. Bd3 d5

This is all part of the opening. White is given a slight advantage by **Stockfish**.

Figure 2.8 tells us Caruana successfully seized territorial control (Space) in the opening up to move 15. In Figure 2.9, look at how the Mobility values drop after the unusual early queen trade on move 8.

**Position 2: After 15. ... d5**



After 15. ... d5.

### 2.2.3 The early middlegame (16-25)

The game continued:

16. Nc2! Bd6 17. Nxe6 Bxe6 18. Kf2 h5 19. h4 Nc8 20. Ne3 Ne7 21. g3 c5!  
22. Bc2 O-O 23. Rd1 Rfd8 24. Ng2 cxd4! 25. cxd4! Rac8

This ends the early middlegame. The game is even, according to Stockfish.

**Position 3:** After 25. ... Rac8

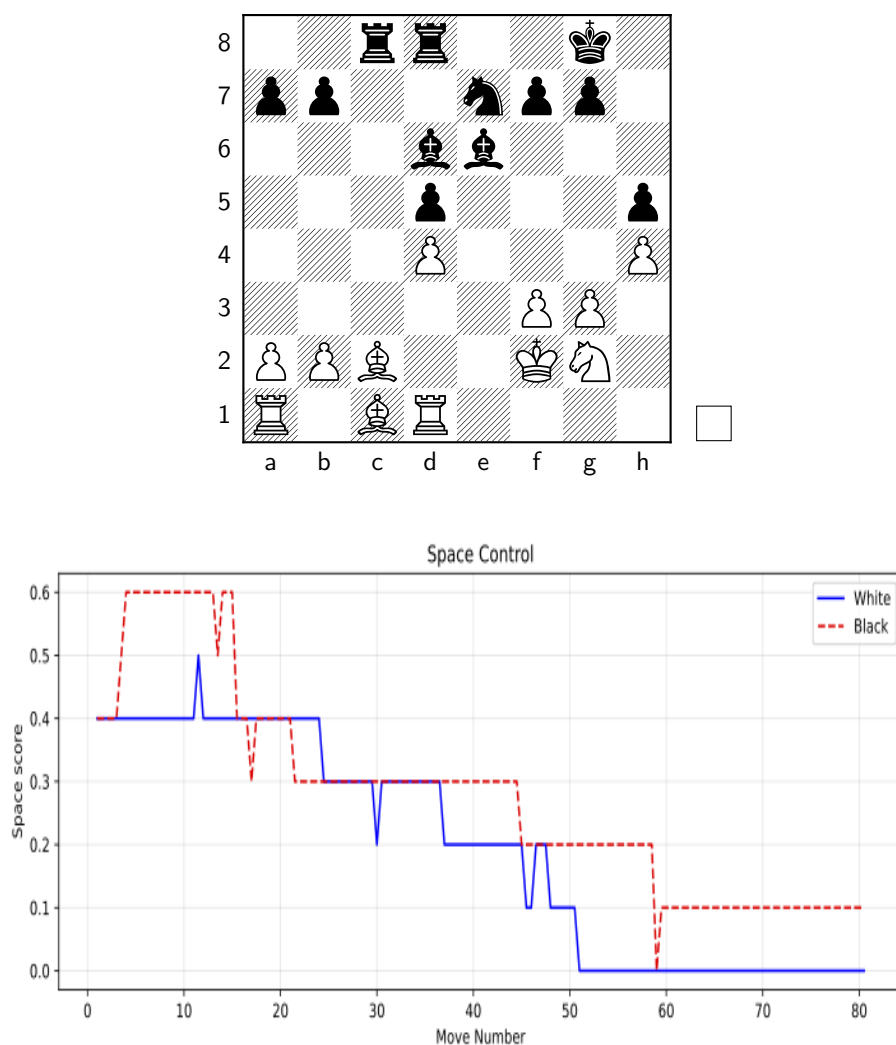


Figure 2.8: (top) Position after 25 ... Rac8 in Carlsen-Caruana, Game 6 (2018). (bottom) The Space metric for that game.

### 2.2.4 The later middlegame (26-33)

The game continued:

**26. Bb3! Nc6 27. Bf4! Na5 28. Rdc1! Bb4 29. Bd1 Nc4! 30. b3! Na3 31. Rxc8! Rxc8! 32. Rc1 Nb5 33. Rxc8+! Bxc8!**

**Position 4: After 33. ... Bxc8!**

Note how the Mobility metric drops after these piece exchanges.



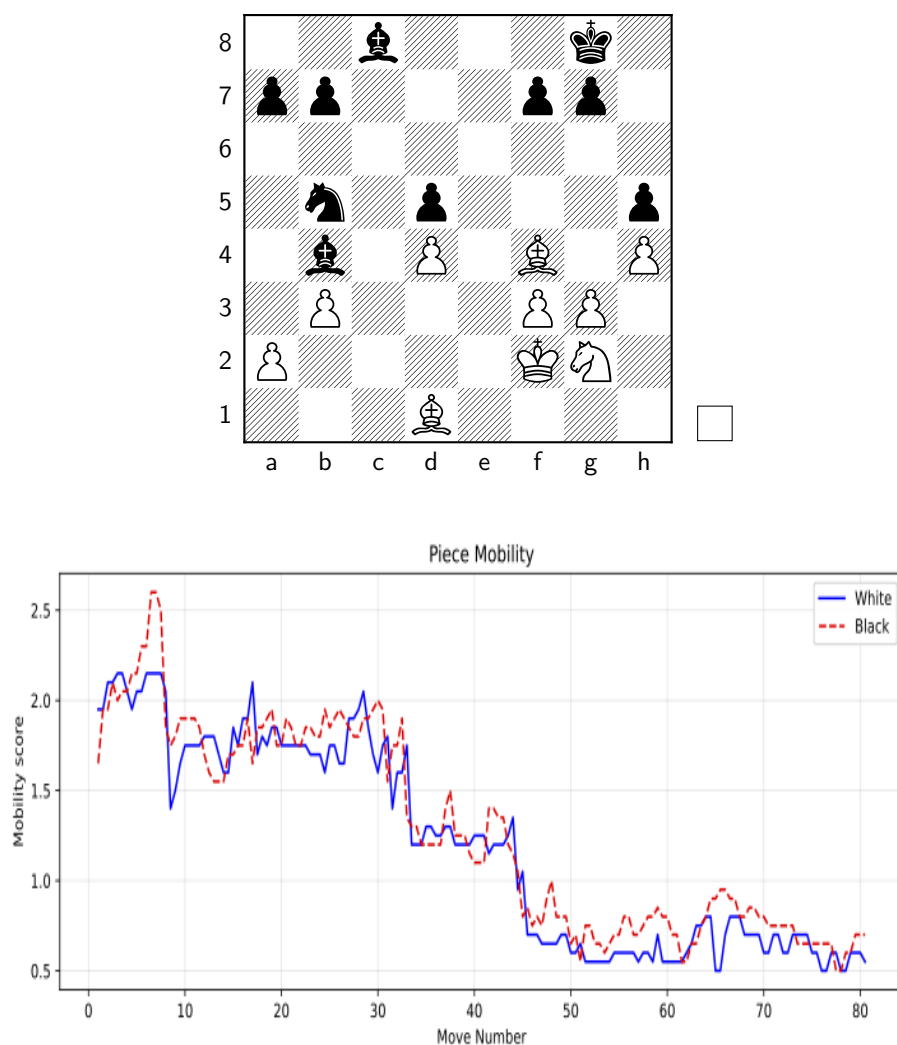


Figure 2.9: (top) Position after 33 ... Bxc8 in Carlsen-Caruana, Game 6 (2018). (bottom) The Mobility metric for that game.

### 2.2.5 The early endgame (34-67)

The game continued:

**34. Ne3 Nc3! 35. Bc2 Ba3 36. Bb8 a6 37. f4 Bd7 38. f5 Bc6 39. Bd1! Bb2 40. Bxh5 Ne4+ 41. Kg2 Bxd4! 42. Bf4 Bc5 43. Bf3! Nd2!**

This is a very interesting move, setting in motion a long combination, dropping piece Mobility but escalating Threats.

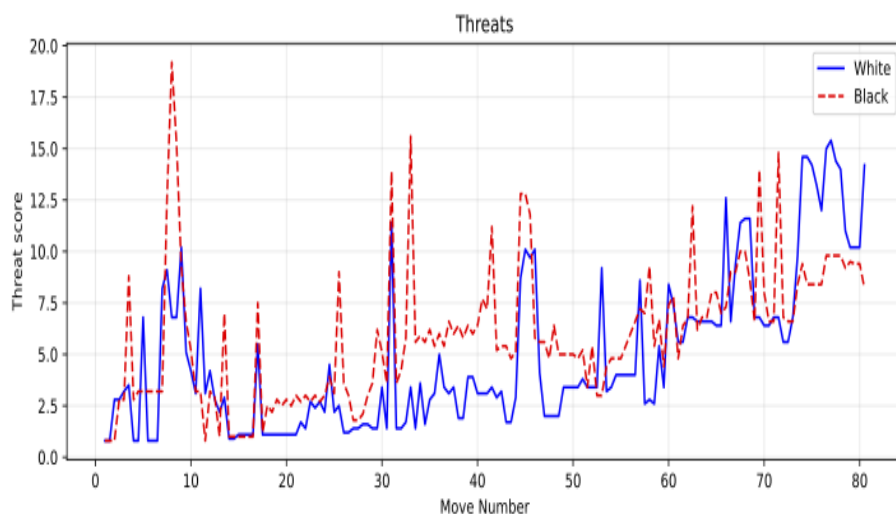
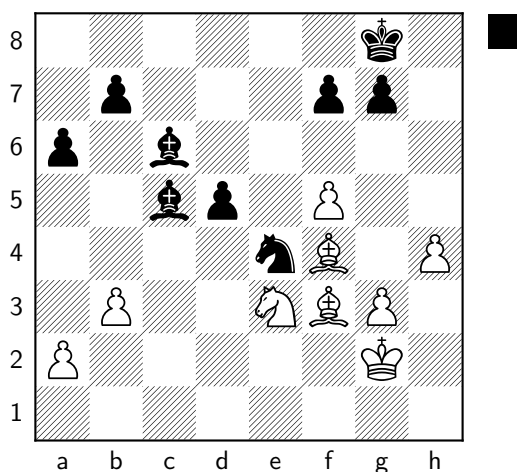
**Position 5: After 43. Bf3**

Figure 2.10: (top) Position after 43 Bf3 and before 43 ... Nd2! in Carlsen-Caruana, Game 6 (2018). (bottom) The Threats metric for that game.

**44. Bxd5?**

Stockfish didn't like Carlsen's move and suggest to consider instead Nd1 or Nf1 or Nc2.

**Position 6: After 44. Bxd5**

Evaluation: -0.81

The game continued:

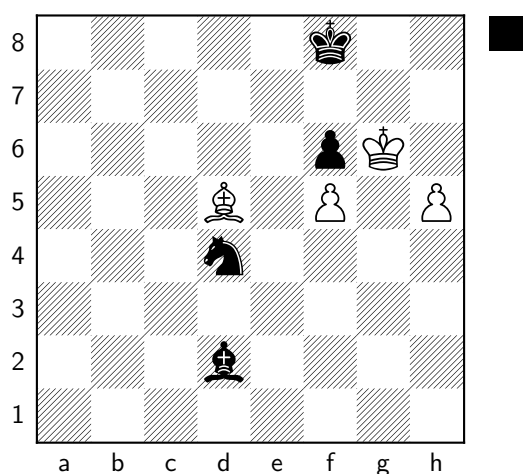
44...Bxe3! 45. Bxc6! Bxf4! 46. Bxb7! Bd6! 47. Bxa6 Ne4! 48. g4! Ba3! 49. Bc4 Kf8 50. g5 Nc3! 51. b4! Bxb4 52. Kf3 Na4! 53. Bb5 Nc5?! 54. a4?! f6 55.

**Kg4 Ne4 56. Kh5 Be1! 57. Bd3 Nd6 58. a5 Bxa5 59. gxf6! gxf6! 60. Kg6! Bd8 61. Kh7! Nf7 62. Bc4 Ne5! 63. Bd5 Ba5 64. h5! Bd2 65. Ba2 Nf3 66. Bd5 Nd4! 67. Kg6??**

Stockfish considered this to be a losing mistake, suggesting instead: Be4.

### Position 7: After 67. Kg6

Evaluation: -2.47



### 2.2.6 The later endgame (67-80)

Black has a winning advantage at this point, according to Stockfish.

**67...Bg5! 68. Bc4 Nf3??**

This is, according to Stockfish a blunder, recommending instead Bh4. According to Stockfish, with depth=23, with this knight move Caruana missed a win.

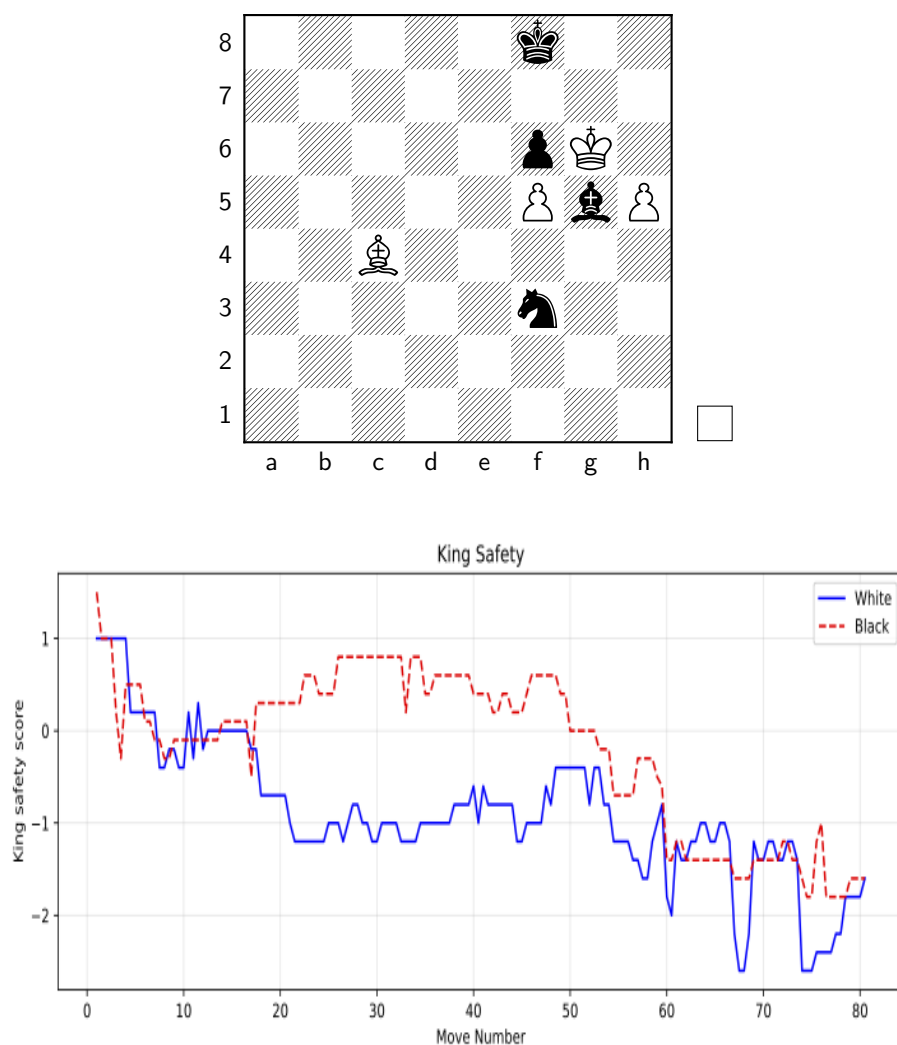


Figure 2.11: (top) Position after 68 ... Nf3?? (moving from d4) in Carlsen-Caruaana, Game 6 (2018). (bottom) The King Safety metric for that game.

After 18 Kf2, the White king enters the game early instead of castling, which here created the opportunity for Black to make more threats. Based on Figure 2.11 (bottom), note the King Safety line drops into negative territory (-0.7 to -1.2) after the opening and stays there. On the other hand, Black's King Safety was positive from moves 18 to 50 (after 49 ... Kf8). Since Carlsen's king was more exposed to checks and needed protection, Caruana was able to keep stronger positional evaluations, even with equal material.

The game concluded this way:

**69. Kh7! Ne5 70. Bb3 Ng4 71. Bc4 Ne3! 72. Bd3! Ng4! 73. Bc4! Nh6 74. Kg6! Ke7 75. Bb3! Kd6 76. Bc2! Ke5 77. Bd3 Kf4 78. Bc2! Ng4! 79. Bb3 Ne3 80. h6! Bxh6! 1/2-1/2**

### 2.2.7 FTI summary

The various types of FTIs are designed to characterize the type of play. Recall that the FTI metric is a linear combination of the four positional metrics, Space (S), Mobility (M), King Safety (KS), and Threats (T):

$$FTI = w_S S + w_M M + w_{KS} KS + w_T T.$$

Think of the FTI as a coordinated pressure gauge. Using it to predict a game's outcome requires a player to hold positional dominance for 10 consecutive 'plies' (half-moves) to predict a win. For the Carlsen-Caruana match, predicting the outcome (win/loss/draw) using Fireteam Indices is a struggle because the game was so well-played that the advantages according to most metrics were small.

In this case, CGA was run over all 12 games Carlsen and Caruana played for the 2018 World Chess championships. The CGA program computes the weights  $(w_S, w_M, w_{KS}, w_T)$  such that, over all those games, the FTI with those "signature" weights predicts the outcome more successfully than any other choices. The outcome was

$$(w_S, w_M, w_{KS}, w_T) = (0.0, 0.0, 0.0, 1.0),$$

with  $R^2 = 0$  (indicating a failure of linear regression to find a line of "best" fit).

Either by looking at the plots for FTI1 and FTI5, we see CGA suggests Caruana should have won the game and plays with a solid, harmonious style.

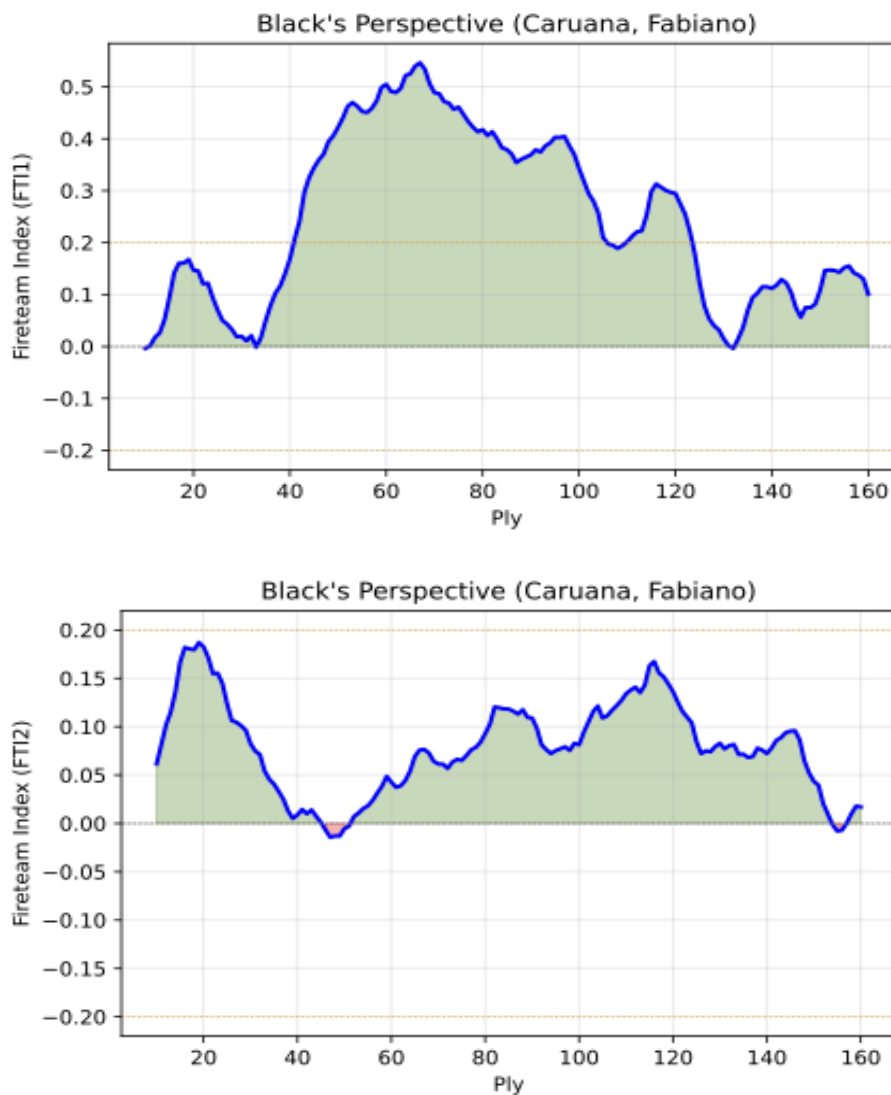


Figure 2.12: Both plots are for game 6 of Carlsen-Caruana, 2018. (top) Fireteam index, FTI1 (harmonious). (bottom) Fireteam index, FTI2 (tactical). The second plot never exceeds the confidence threshold of  $\pm 0.2$ .

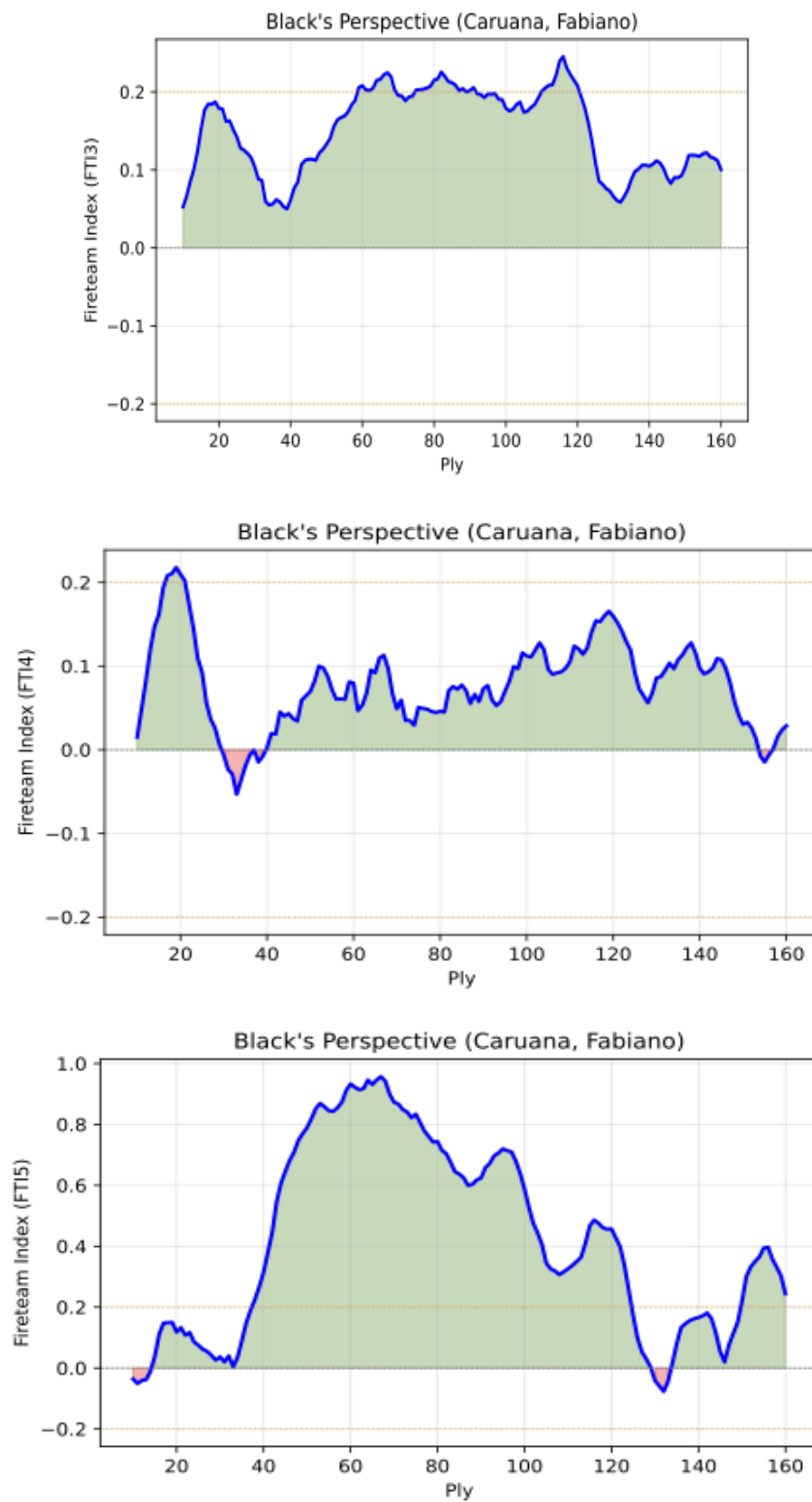


Figure 2.13: These plots are also for game 6 of Carlsen-Caruana, 2018. (top) Fireteam index, FTI3 (strategic). (middle) Fireteam index, FTI4 (dynamic). (bottom) Fireteam index, FTI5 (solid).

Of all the FTI plots in Figures 2.12 and 2.13, the first and last one most strongly suggests a win for Black. The other FTIs also suggest a Black win, but with less confidence.



# Chapter 3

## Appendix: Chess Game Analyzer module reference

This is the readme file for the CGA program [CGA26] generated by `claude` (Anthropic), with some editing by me.

### 3.1 Overview

The Chess Game Analyzer is a Python module for deep positional analysis of chess games using Stockfish, with comprehensive L<sup>A</sup>T<sub>E</sub>X report generation and win prediction metrics.

Chess Game Analyzer goes beyond simple engine evaluation to provide interpretable positional metrics for every move of a chess game. It computes Space control, Mobility, King Safety, and Threats directly from board positions, then combines these into **Fireteam Index (FTI)** metrics that can predict game outcomes based on sustained positional advantages.

The module generates professional L<sup>A</sup>T<sub>E</sub>X reports with annotated games, critical position diagrams, matplotlib plots, and detailed methodology explanations—suitable for chess study, coaching, or research.

### 3.2 Background for Newcomers

#### 3.2.1 What is Stockfish?

**Stockfish**<sup>1</sup> is the world's strongest open-source chess engine. It analyzes positions and calculates the best moves by searching millions of positions per second. This module uses Stockfish to evaluate positions, but computes the positional breakdown (Space, Mobility, etc.) directly using the `python-chess` library.

**Stockfish must be installed on your system.** Download it from the official website and note the installation path.

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<sup>1</sup><https://stockfishchess.org/>

### 3.2.2 What is a Centipawn?

Chess engines measure advantages in **centipawns (cp)**—hundredths of a pawn. An evaluation of +100 means White is ahead by roughly one pawn’s worth of material or positional compensation. An evaluation of −250 means Black has about a 2.5 pawn advantage.

| Evaluation             | Meaning             |
|------------------------|---------------------|
| 0 to $\pm 50$          | Roughly equal       |
| $\pm 50$ to $\pm 150$  | Slight advantage    |
| $\pm 150$ to $\pm 300$ | Clear advantage     |
| $\pm 300$ to $\pm 500$ | Winning advantage   |
| $\pm 500+$             | Decisive advantage  |
| $\pm 10000$            | Checkmate is forced |

### 3.2.3 Python Requirements

The module requires:

- **Python 3.8 or higher**
- **python-chess**: The chess library that handles board representation, move generation, and PGN parsing
- **matplotlib** (optional): For generating plots; the module works without it using ASCII plots

To install the dependencies:

```
pip install python-chess matplotlib
```

## 3.3 Installation

1. Install Stockfish from <https://stockfishchess.org/download/>
2. Clone the repository or download `chess_game_analyzer.py`
3. Install Python dependencies:

```
pip install python-chess matplotlib
```

4. Verify Stockfish path:

```
# Linux/Mac - find where Stockfish is installed
which stockfish
```

```
# Or test directly
stockfish
```

## 3.4 Quick Start

### 3.4.1 Command Line

```
# Analyze a single game
python chess_game_analyzer.py game.pgn -o analysis.tex

# Analyze with custom Stockfish path and deeper analysis
python chess_game_analyzer.py game.pgn -o analysis.tex \
    --stockfish /usr/local/bin/stockfish \
    --depth 22 \
    --time 2.0

# Analyze multiple games as a book
python chess_game_analyzer.py tournament.pgn -o book.tex --book \
    --book-title "My Tournament Games" \
    --book-author "Your Name"
```

### 3.4.2 Python API

```
from chess_game_analyzer import (
    EnhancedGameAnalyzer,
    EnhancedLaTeXReportGenerator,
    classify_game_character
)

# Analyze a game
with EnhancedGameAnalyzer("/usr/local/bin/stockfish",
                          depth=20, time_limit=1.0) as analyzer:
    result = analyzer.analyze_game("game.pgn")

# Access move-by-move data
for move in result.moves:
    pos = move.positional_eval
    print(f"Move {move.ply}: {move.move_san}")
    print(f"  Space: W={pos.space_white:.2f}, B={pos.space_black:.2f}")
    print(f"  Mobility: W={pos.mobility_white:.2f}, B={pos.mobility_black:.2f}")
    print(f"  Eval: {move.eval_after/100:+.2f} pawns")

# Generate LaTeX report
latex = EnhancedLaTeXReportGenerator.generate_report(
    result,
    include_plots=True,
    include_methodology=True
)
```

```
with open("analysis.tex", "w") as f:
    f.write(latex)
```

## 3.5 Positional Metrics Explained

### 3.5.1 Space

**Space** measures territorial control, particularly in the center and the opponent's half of the board.

The computation counts squares on central files (c–f) in ranks 2–4 (for White) or 5–7 (for Black) that are:

1. Attacked by at least one friendly pawn, *and*
2. Not occupied by an enemy piece

The raw count is divided by 10 to normalize the metric. Values typically range from 0.1 to 0.7.

**Example from Estrin vs Berliner (1965), move 20 (10...e4):**

- White Space: 0.30
- Black Space: 0.30

At this point in a sharp Two Knights Defense, space is roughly equal, but the position is about to explode tactically.

### 3.5.2 Mobility

**Mobility** counts the number of safe squares available to each piece, weighted by piece type.

“Safe” squares exclude:

- Squares attacked by enemy pawns
- Squares that would result in material loss through exchange

Mobility in enemy territory is weighted higher. The metric captures piece activity and coordination.

**Example from Estrin vs Berliner, move 30 (15...Bxf3):**

- White Mobility: 1.95
- Black Mobility: 2.5

After White's blunder 15.Be2??, Black's pieces have become significantly more active.

### 3.5.3 King Safety

**King Safety** combines several factors:

1. **Pawn shield strength:** Pawns on the 2nd/3rd rank in front of the castled king
2. **King geometry:** Distance of enemy pieces to your king
3. **Attack units:** Enemy pieces attacking squares near your king

Positive values indicate a safe king; negative values indicate danger.

**Example 3.5.1.** *From Estrin vs Berliner, move 52 (26...hxg2+):*

- *White King Safety:*  $-1.40$
- *Black King Safety:*  $-0.70$

*White's king is significantly more exposed after the pawn storm breaks through.*

### 3.5.4 Threats

**Threats** evaluates the tactical tension in the position:

- **Hanging pieces:** Pieces attacked but not defended (weighted by piece value  $\times 2$ )
- **Attacks by lower-value pieces:** e.g., a knight attacking a queen
- **King zone pressure:** Attacks on squares near the enemy king
- **Safe checks:** Available checking moves that don't lose material
- **Weak squares (holes):** Squares that can never be defended by pawns

Raw Threats values typically range from 1 to 30. In the Fireteam Index calculations, Threats is scaled down (divided by 20) to be comparable with the other metrics.

**Example 3.5.2.** *From Estrin vs Berliner, move 24.Nc4: The Threats graph has spikes and valleys in this part of the game.*

- *White Threats:*  $21.80$
- *Black Threats:*  $8.50$

*White has significant pressure, but Black's position is resilient.*

## 3.6 The Fireteam Index (FTI)

### 3.6.1 Background: Chess Metrics and Win Prediction

Traditional chess analysis focuses on **centipawn evaluation**—a single number indicating who stands better. But this doesn’t tell you *why* someone is winning or *how* the advantage manifests.

The **Fireteam Index** is inspired by Rithmomachia,<sup>2</sup> a medieval mathematical board game where one victory condition is establishing a “well-coordinated fireteam” of pieces deep in enemy territory forming arithmetic, geometric, or harmonic progressions.

Similarly, in chess, sustained positional dominance across multiple factors—space, activity, king safety, and threats—often precedes victory. The FTI combines these factors into a single metric that tracks this multi-dimensional advantage over time.

### 3.6.2 FTI Signatures

Different players have different styles. A tactical player might win through direct threats, while a positional player grinds down opponents through space and piece activity. The module provides five FTI “signatures” with different weightings:

| FTI Variant                   | Space | Mobility | King Safety | Threats | Style              |
|-------------------------------|-------|----------|-------------|---------|--------------------|
| FTI <sub>1</sub> (Harmonious) | 0.25  | 0.25     | 0.25        | 0.25    | Balanced           |
| FTI <sub>2</sub> (Tactical)   | 0.60  | 0.10     | 0.00        | 0.30    | Space + threats    |
| FTI <sub>3</sub> (Strategic)  | 0.70  | 0.10     | 0.10        | 0.10    | Spatial domination |
| FTI <sub>4</sub> (Dynamic)    | 0.20  | 0.60     | 0.00        | 0.20    | Piece activity     |
| FTI <sub>5</sub> (Solid)      | 0.30  | 0.20     | 0.50        | 0.00    | Defensive play     |

The FTI formula for a given player is:

$$\text{FTI} = w_S \cdot \Delta S + w_M \cdot \Delta M + w_K \cdot \Delta K + w_T \cdot \Delta T' \quad (3.1)$$

where each  $\Delta$  is (Player’s value – Opponent’s value) and  $T' = T/20$  (scaled for comparability).

### 3.6.3 Prediction Algorithms

Three algorithms use the FTI to predict game outcomes:

#### 1. Per-Ply (Raw).

- Examines raw FTI at each ply after move 8 (ply 16)
- Predicts *win* if  $\text{FTI} > \epsilon$  (default 0.2) for 10 consecutive plies
- Captures sudden, decisive advantages

<sup>2</sup>Rithmomachia, also known as the “philosopher’s game,” was a medieval mathematical board game popular in European monasteries and universities from the 11th to 16th centuries.

**2. Windowed (Smoothed).**

- Computes a 10-ply rolling average of FTI
- Applies the same streak detection to smoothed values
- Filters out tactical noise; captures sustained trends

**3. Balance (Confidence-based).**

- Counts plies where  $\text{FTI} > \epsilon$  versus  $\text{FTI} < -\epsilon$
- Measures volatility (sign changes)
- Computes confidence =  $|B_n| \times (1 - V/V_{\max})$
- Predicts *win/loss* if confidence  $> 0.15$ ; otherwise *draw*.
- Best at detecting drawn games and balanced struggles.

**3.7 Concrete Example: Estrin vs Berliner (1965)**

Let us walk through the analysis of a famous game from the 5th World Correspondence Chess Championship—**Yakov Estrin vs Hans Berliner**, featuring the sharp Two Knights Defense.

**3.7.1 The Game (C57 – Two Knights Defense)**

As we know from the previous chapter, the first few moves of the game are as follows.

**1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6 4.Ng5 d5 5.exd5 b5!?**

Berliner plays the Ulvestad Variation (5...b5), a sharp gambit that had been analyzed deeply in correspondence play.

**3.7.2 Analysis Output****Player Statistics.**

| Metric             | Estrin (White) | Berliner (Black) |
|--------------------|----------------|------------------|
| Accuracy           | 87.5%          | 92.1%            |
| Avg centipawn loss | 26.6           | 16.5             |
| Best moves         | 18             | 21               |
| Blunders           | 2              | 0                |

**Game Character Classification.**

- Evaluation range:  $m_1 = -7.04$ ,  $m_2 = +2.24$
- Spread:  $d = 9.28$  pawns
- Classification: **Chaotic seesaw battle**

The game swung wildly—White had +2.24 at one point, but Black eventually achieved  $-7.04$ .

**3.7.3 Positional Metrics Summary**

| Metric      | White Average | Black Average |
|-------------|---------------|---------------|
| Space       | 0.30          | 0.22          |
| Mobility    | 1.51          | 1.67          |
| King Safety | $-0.25$       | $-0.22$       |
| Threats     | 6.35          | 5.91          |

Despite the sharp tactical nature, Berliner maintained slightly better piece activity throughout.

**3.7.4 Move-by-Move Data Extract**

The module stores raw positional data in a machine-readable format:

```
% ply, SAN, eval_cp, space_w, space_b, mob_w, mob_b, ks_w, ks_b,
%      threats_w, threats_b
% 29, Be2, -126, 0.300, 0.300, 5.000, 4.600, 0.000, 0.100, 3.800, 4.700
% 30, Bxf3, -204, 0.200, 0.300, 3.900, 5.000, -0.200, 0.500, 7.600, 6.000
% 31, Bxf3, -237, 0.300, 0.300, 4.000, 4.400, 0.200, 0.500, 7.400, 5.200
```

This data can be parsed for further computational analysis, machine learning, or statistical studies.

**3.8 Command Line Interface**

```
python chess_game_analyzer.py [PGN_FILE] [OPTIONS]
```



### 3.8.1 Core Options

| Option               | Description                                      | Default              |
|----------------------|--------------------------------------------------|----------------------|
| -o, --output FILE    | Output L <sup>A</sup> T <sub>E</sub> X file path | stdout               |
| --json-output FILE   | Output raw analysis as JSON                      | none                 |
| -s, --stockfish PATH | Path to Stockfish executable                     | /usr/games/stockfish |
| -d, --depth N        | Analysis depth (higher = slower)                 | 20                   |
| -t, --time SECS      | Time limit per position                          | 1.0                  |
| -q, --quiet          | Suppress progress messages                       | false                |

### 3.8.2 Report Options

| Option           | Description                                    |
|------------------|------------------------------------------------|
| --no-diagrams    | Don't include chess board diagrams             |
| --no-methodology | Don't include methodology explanation section  |
| --no-plots       | Don't include matplotlib plots                 |
| --ascii-plots    | Include ASCII plots (works without matplotlib) |
| --plot-dir DIR   | Directory for plot output files                |

### 3.8.3 Fireteam Index Prediction (Single Game)

| Option                 | Description                                       |
|------------------------|---------------------------------------------------|
| --prediction W B       | Include FTI prediction for White (W) or Black (B) |
| --prediction-name NAME | Player name for display (e.g., "Berliner")        |

### 3.8.4 Book Mode (Multiple Games)

| Option                 | Description                                    |
|------------------------|------------------------------------------------|
| --book                 | Analyze all games in PGN as a book             |
| --book-title TITLE     | Title for the book                             |
| --book-author AUTHOR   | Author name                                    |
| --prediction-winner    | Track winner in decisive games (default: true) |
| --prediction-name NAME | Track specific player by name                  |
| --no-prediction        | Disable FTI prediction section entirely        |

### 3.8.5 Examples

```
# Deep analysis with plots
python chess_game_analyzer.py game.pgn -o report.tex \
  --stockfish /opt/stockfish/stockfish \
  --depth 24 --time 3.0
```

```
# Quick analysis without graphics
python chess_game_analyzer.py game.pgn -o report.tex \
    --depth 16 --time 0.5 --no-plots --ascii-plots

# Analyze Berliner's games as a collection
python chess_game_analyzer.py berliner_games.pgn -o berliner_book.tex \
    --book \
    --book-title "Berliner's World Championship Games" \
    --book-author "Analysis by Stockfish 17" \
    --prediction-name "Berliner"

# JSON output for programmatic access
python chess_game_analyzer.py game.pgn --json-output analysis.json
```

## 3.9 Python API

### 3.9.1 Main Classes

**EnhancedGameAnalyzer.** The core analysis engine. Use as a context manager to ensure proper cleanup.

```
with EnhancedGameAnalyzer(
    stockfish_path="/usr/local/bin/stockfish",
    depth=20,          # Search depth (plies)
    time_limit=1.0,    # Seconds per position
    extract_positional=True # Compute Space/Mobility/etc.
) as analyzer:
    result = analyzer.analyze_game("game.pgn")
    # or analyze_all_games() for multi-game PGNs
```

**EnhancedGameAnalysisResult.** Contains all analysis data:

- **moves:** List of **EnhancedMoveAnalysis** objects
- **white\_stats, black\_stats:** Accuracy and move classification counts
- **brilliant\_sacrifices:** Detected sacrifices
- **critical\_positions:** Key moments for diagrams
- **game\_character:** Classification (balanced/tactical/chaotic, etc.)
- **positional\_summary:** Aggregated Space/Mobility/etc. statistics

**PositionalEvaluation.** Attached to each move, containing:

- `space_white`, `space_black`
- `mobility_white`, `mobility_black`
- `king_safety_white`, `king_safety_black`
- `threats_white`, `threats_black`

### 3.9.2 Utility Functions

```
from chess_game_analyzer import (
    parse_raw_positional_data,
    compute_fireteam_index,
    predict_outcome_per_ply,
    predict_outcome_windowed,
    predict_outcome_balance,
    classify_game_character
)

# Parse raw data from LaTeX file
data = parse_raw_positional_data(tex_content, game_number=7)

# Compute FTI from positional data
fti_data = compute_fireteam_index(data)

# Run predictions with custom weights
from chess_game_analyzer import FTI2_WEIGHTS # Tactical signature

result = predict_outcome_per_ply(
    data,
    player_color="B",
    player_name="Berliner",
    weights=FTI2_WEIGHTS,
    cutoff_ply=16,
    streak_length=10,
    epsilon=0.2
)
print(f"Prediction: {result.prediction}")
print(f"Max streak: {result.max_streak} plies")
```

## 3.10 Parameter Reference

### 3.10.1 Analysis Parameters

| Parameter           | Type  | Default              | Description                          |
|---------------------|-------|----------------------|--------------------------------------|
| stockfish_path      | str   | /usr/games/stockfish | Path to Stockfish binary             |
| depth               | int   | 20                   | Search depth in plies                |
| time_limit          | float | 1.0                  | Maximum seconds per position         |
| min_diagram_spacing | int   | 6                    | Min ply distance between diagrams    |
| top_n_swings        | int   | 2                    | “Biggest swing” positions to include |

### 3.10.2 FTI Prediction Parameters

| Parameter            | Type  | Default                  | Description                        |
|----------------------|-------|--------------------------|------------------------------------|
| cutoff_ply           | int   | 16                       | Opening plies to skip              |
| streak_length        | int   | 10                       | Consecutive plies for WIN          |
| epsilon              | float | 0.2                      | Min FTI for “meaningful advantage” |
| window_size          | int   | 10                       | Rolling average window             |
| weights              | tuple | (0.25, 0.25, 0.25, 0.25) | $(w_S, w_M, w_K, w_T)$             |
| v_ratio              | float | 0.20                     | Volatility threshold ratio         |
| confidence_threshold | float | 0.15                     | Min confidence for WIN/LOSS        |

### 3.10.3 Available Weight Presets

```

from chess_game_analyzer import (
    FTI1_WEIGHTS, # (0.25, 0.25, 0.25, 0.25) - Harmonious
    FTI2_WEIGHTS, # (0.60, 0.10, 0.00, 0.30) - Tactical
    FTI3_WEIGHTS, # (0.70, 0.10, 0.10, 0.10) - Strategic
    FTI4_WEIGHTS, # (0.20, 0.60, 0.00, 0.20) - Dynamic
    FTI5_WEIGHTS, # (0.30, 0.20, 0.50, 0.00) - Solid
)

```

### 3.10.4 Move Classification Thresholds

| Classification | Centipawn Loss                    |
|----------------|-----------------------------------|
| Best           | 0 (matches engine’s first choice) |
| Excellent      | 0–10 cp                           |
| Good           | 10–25 cp                          |
| Inaccuracy     | 25–50 cp                          |
| Mistake        | 50–150 cp                         |
| Blunder        | 150+ cp                           |

## 3.11 Output Formats

### 3.11.1 L<sup>A</sup>T<sub>E</sub>X Report

The primary output is a L<sup>A</sup>T<sub>E</sub>X document that can be compiled with `pdflatex` or `xelatex`. It includes:

1. **Game Information:** Event, players, date, opening
2. **Player Statistics:** Accuracy, move counts by classification
3. **Positional Analysis:** Space/Mobility/KingSafety/Threats summaries with plots
4. **Game Character Classification:** Balanced/tactical/chaotic assessment
5. **Annotated Game:** Move-by-move with NAG symbols and comments
6. **Critical Positions:** Diagrams at key moments with best alternatives
7. **Methodology Section:** (optional) Detailed explanation of all metrics
8. **Fireteam Index Prediction:** (optional) Win prediction analysis with plots

Required L<sup>A</sup>T<sub>E</sub>X packages: `xskak`, `chessboard`, `amsmath`, `pgfplots`, `graphicx`, `hyperref`.

### 3.11.2 JSON Output

Raw analysis data in JSON format for programmatic access:

```
{
  "white": "Estrin",
  "black": "Berliner",
  "result": "0-1",
  "moves": [
    {
      "ply": 1,
      "move_san": "e4",
      "eval_after": 15,
      "classification": "best",
      "positional_eval": {
        "space_white": 0.40,
        "space_black": 0.40,
        "mobility_white_mg": 3.90,
        ...
      }
    },
    ...
  ],
}
```

```

    "white_stats": {"accuracy": 87.5, ...},
    "black_stats": {"accuracy": 92.1, ...}
}

```

### 3.11.3 Raw Positional Data

Appended after `\end{document}` in  $\text{\LaTeX}$  files, machine-readable CSV format:

```

% GAME_DATA_START
% 1, e4, 15, 0.400, 0.400, 3.900, 3.300, 1.000, 1.500, 0.800, 0.800
% 2, e5, 24, 0.400, 0.400, 3.900, 3.900, 1.000, 1.000, 0.800, 0.800
...
% GAME_DATA_END

```

Fields: ply, SAN, eval\_cp, space\_w, space\_b, mob\_w, mob\_b, ks\_w, ks\_b, threats\_w, threats\_b.

## 3.12 License

This software is released under your choice of either the **MIT License** or the **Modified BSD License**.

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## 3.13 Acknowledgments

- **Stockfish** (<https://stockfishchess.org/>) — The chess engine powering the evaluations
- **python-chess** (<https://python-chess.readthedocs.io/>) — The excellent chess library
- The concept of the Fireteam Index is inspired by **Rithmomachia**, the medieval “philosopher’s game”

## Chapter 4

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